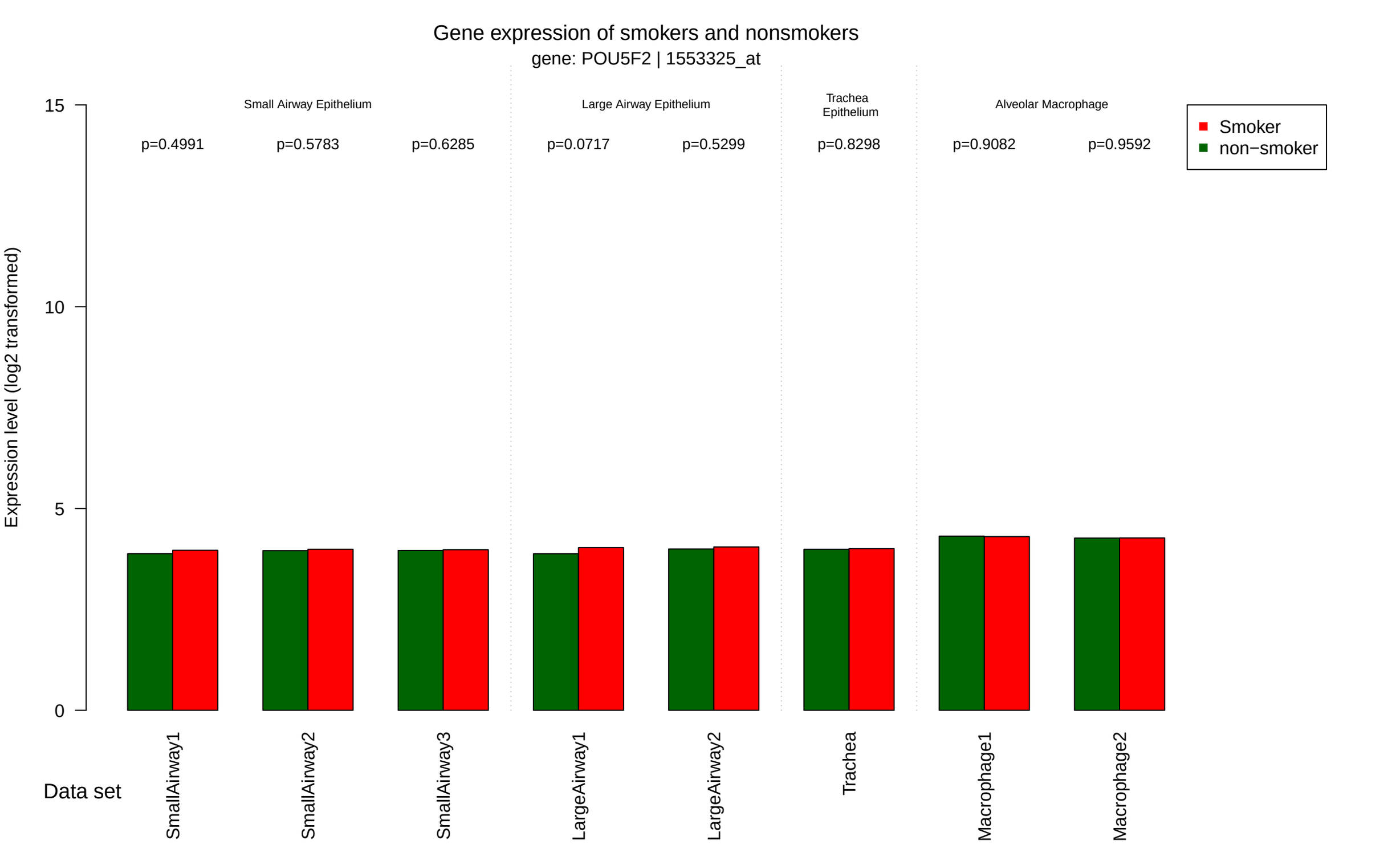
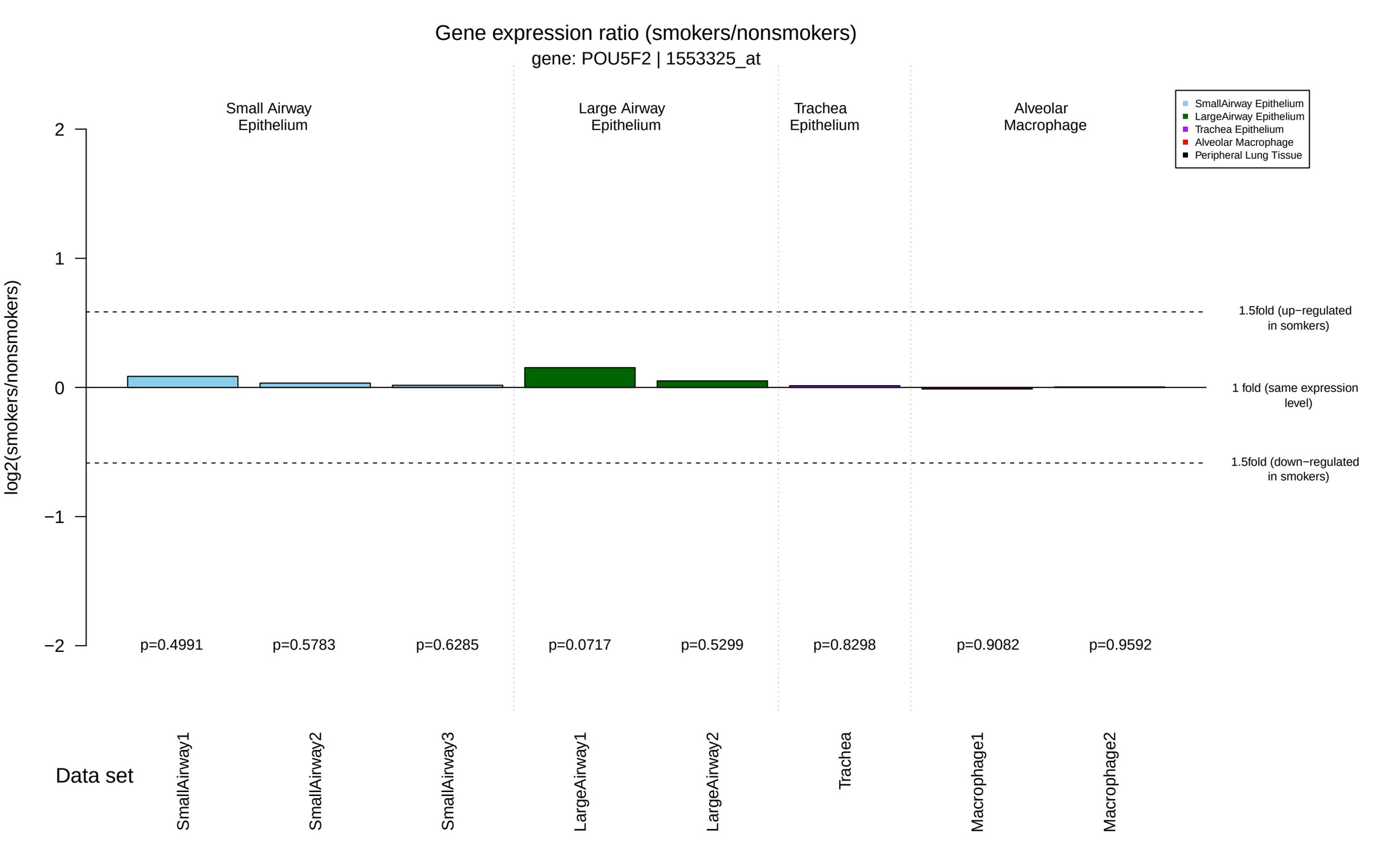
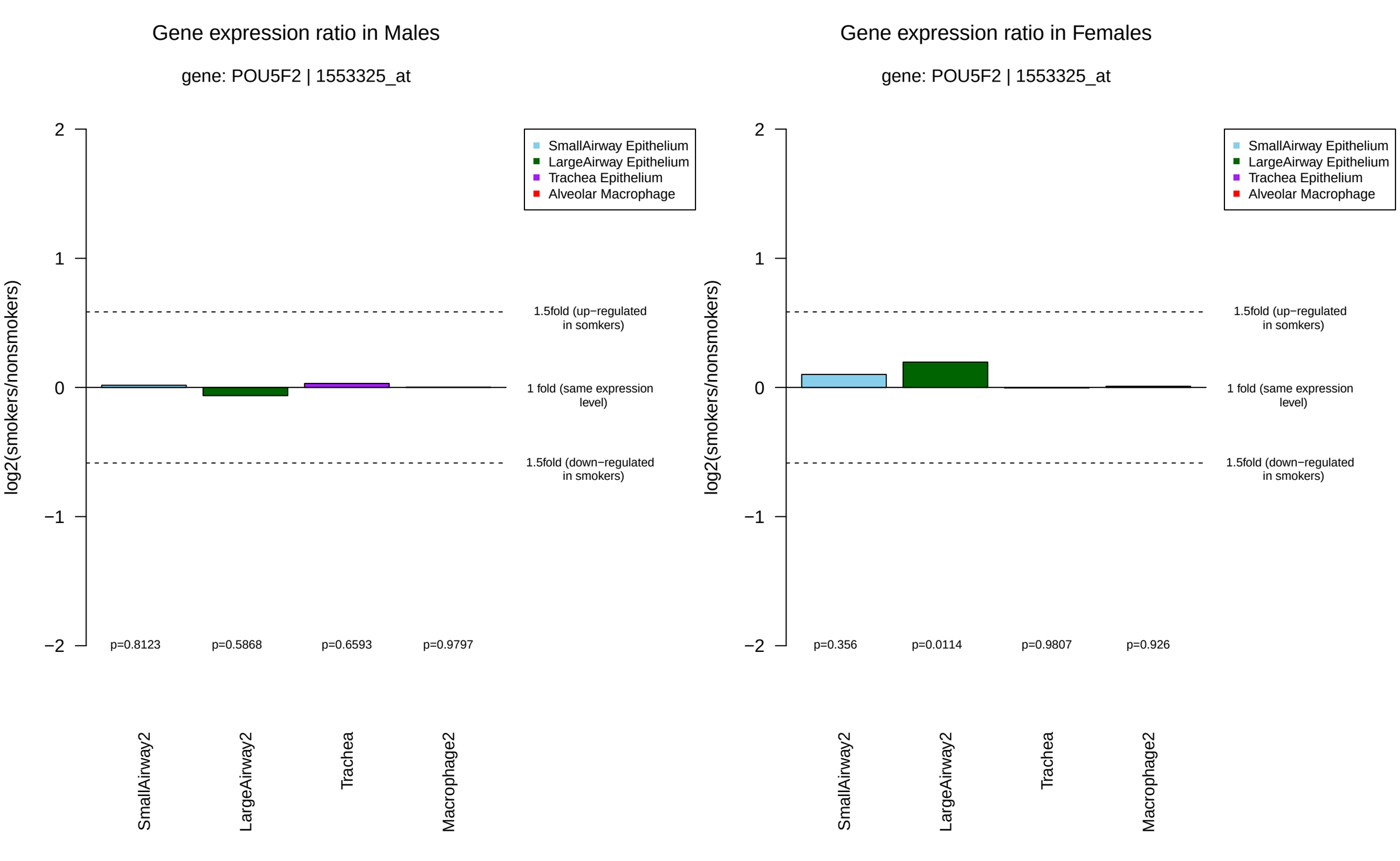
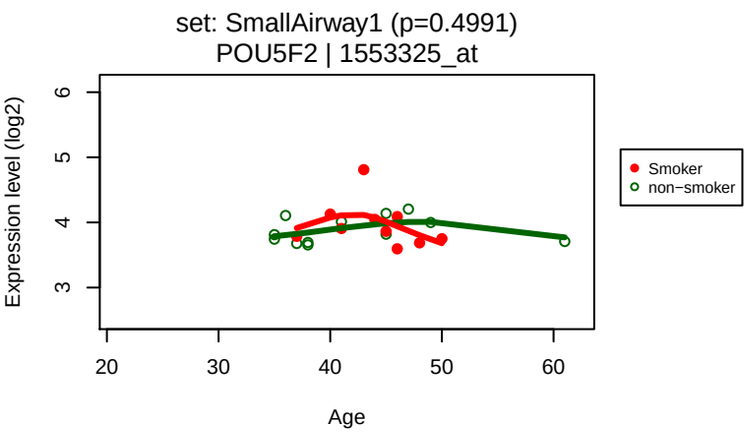
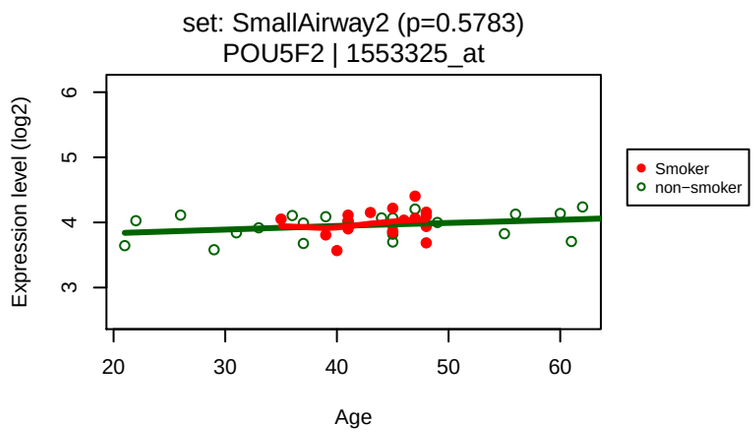
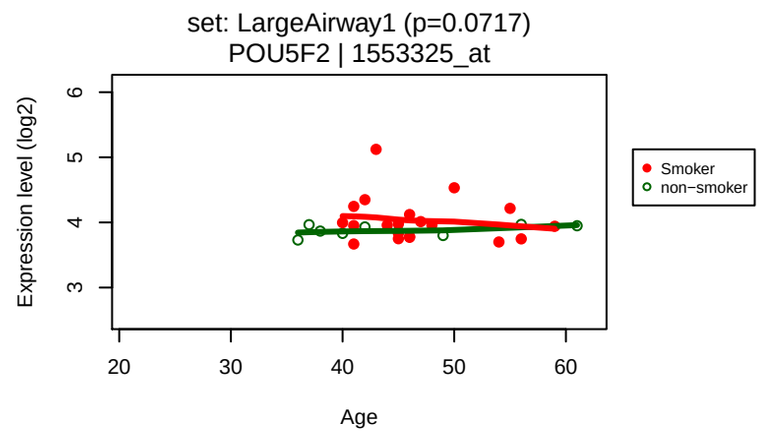

Gene expression of smokers and nonsmokers
gene: POU5F2 | 1553325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4991
p=0.5783
p=0.6285
p=0.0717
p=0.5299
p=0.8298
p=0.9082
p=0.9592
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

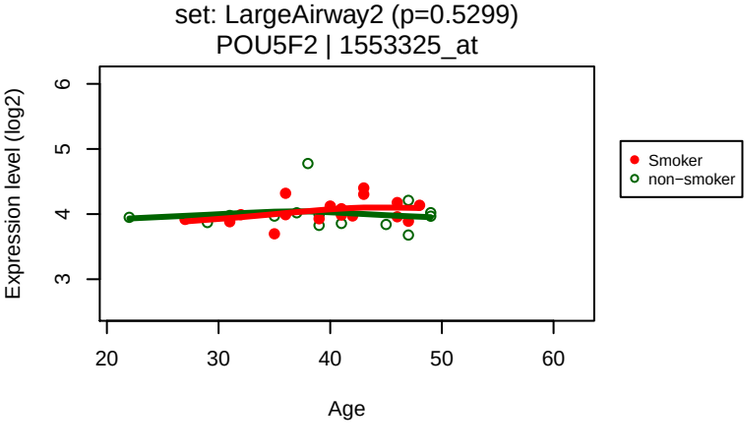

Gene expression ratio (smokers/nonsmokers)
gene: POU5F2 | 1553325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4991
p=0.5783
p=0.6285
p=0.0717
p=0.5299
p=0.8298
p=0.9082
p=0.9592
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

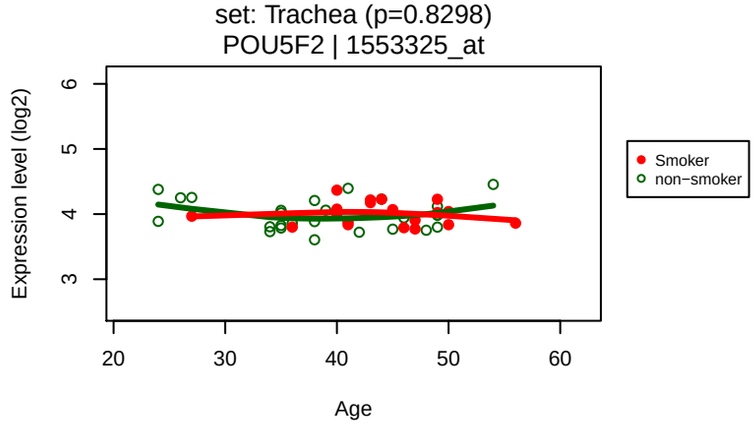

Gene expression ratio in Males
gene: POU5F2 | 1553325_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8123
p=0.5868
p=0.6593
p=0.9797
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POU5F2 | 1553325_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.356
p=0.0114
p=0.9807
p=0.926
SmallAirway2
LargeAirway2
Trachea
Macrophage2

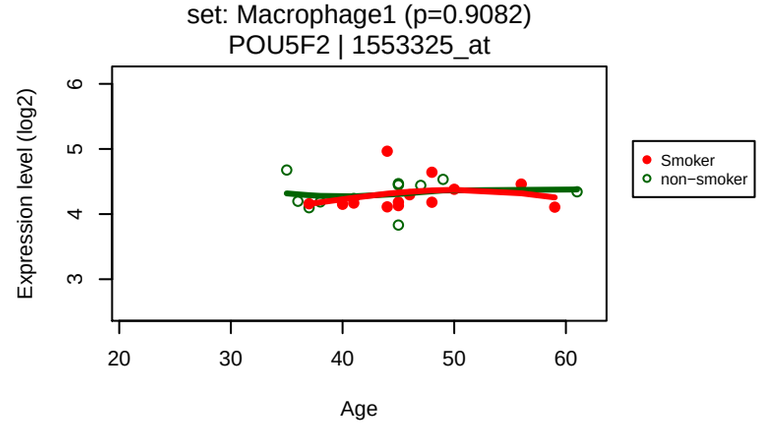

set: SmallAirway1 (p=0.4991)
POU5F2 | 1553325_at
Expression level (log2)
Age
Smoker
non−smoker

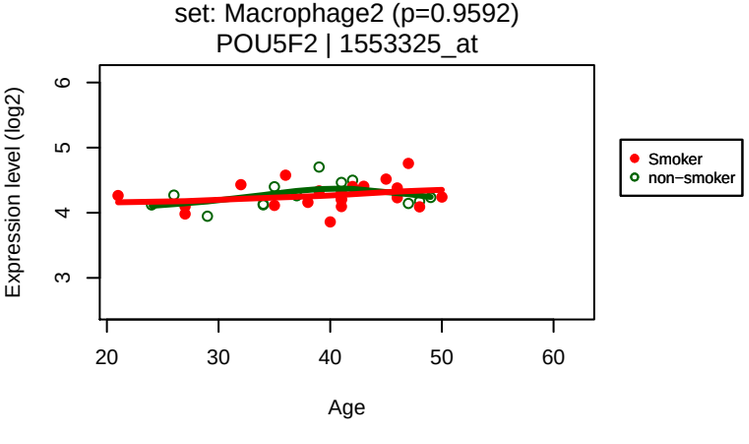

set: SmallAirway2 (p=0.5783)
POU5F2 | 1553325_at
Expression level (log2)
Age
Smoker
non−smoker

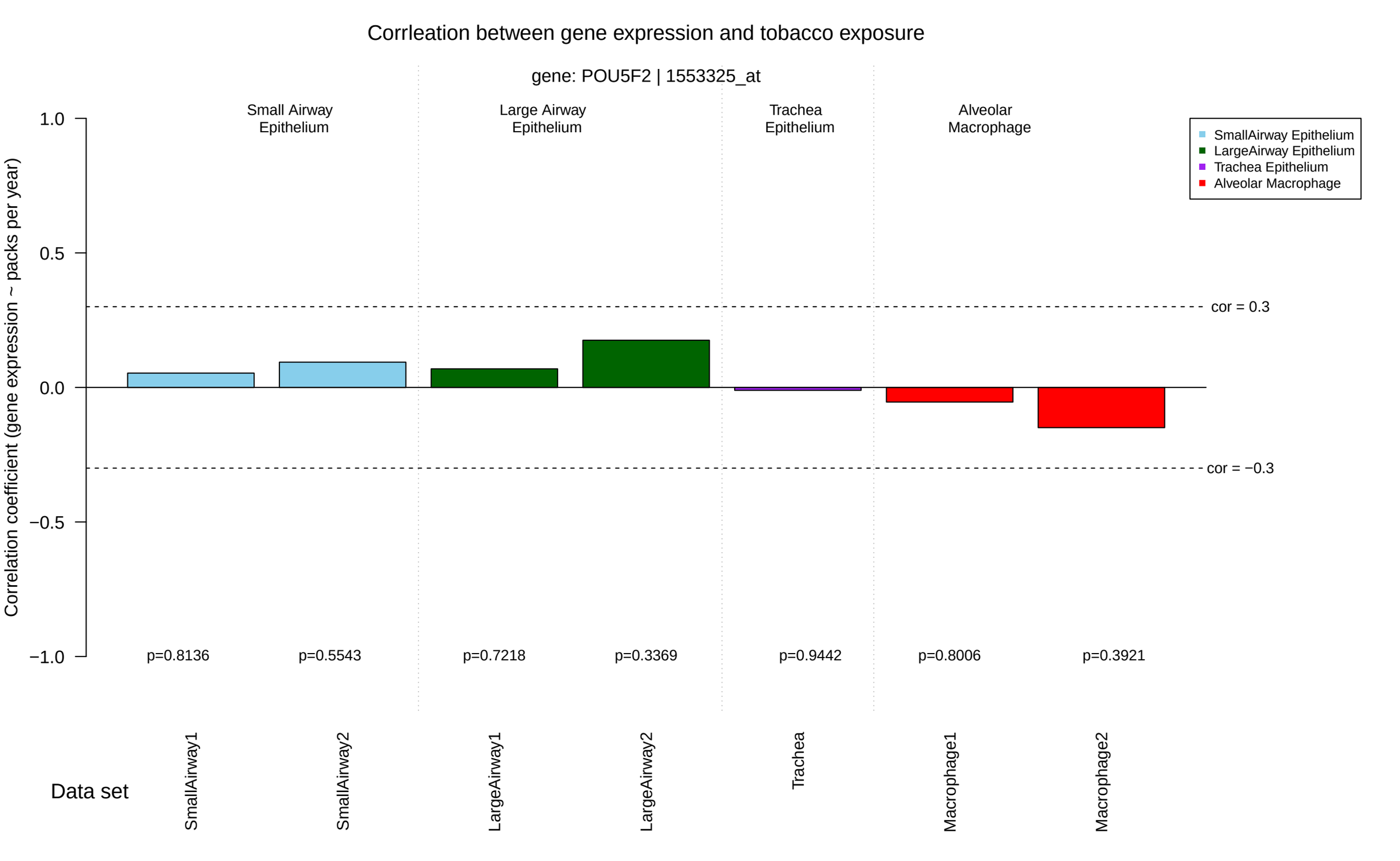

set: LargeAirway1 (p=0.0717)
POU5F2 | 1553325_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5299)
POU5F2 | 1553325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8298)
POU5F2 | 1553325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9082)
POU5F2 | 1553325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9592)
POU5F2 | 1553325_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POU5F2 | 1553325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8136
p=0.5543
p=0.7218
p=0.3369
p=0.9442
p=0.8006
p=0.3921
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2